

Model Quality Report

Holdout performance summary for parametric estimation

The model predicts within 79.8% of actuals for half the holdout, and within 487.0% for nine in ten.

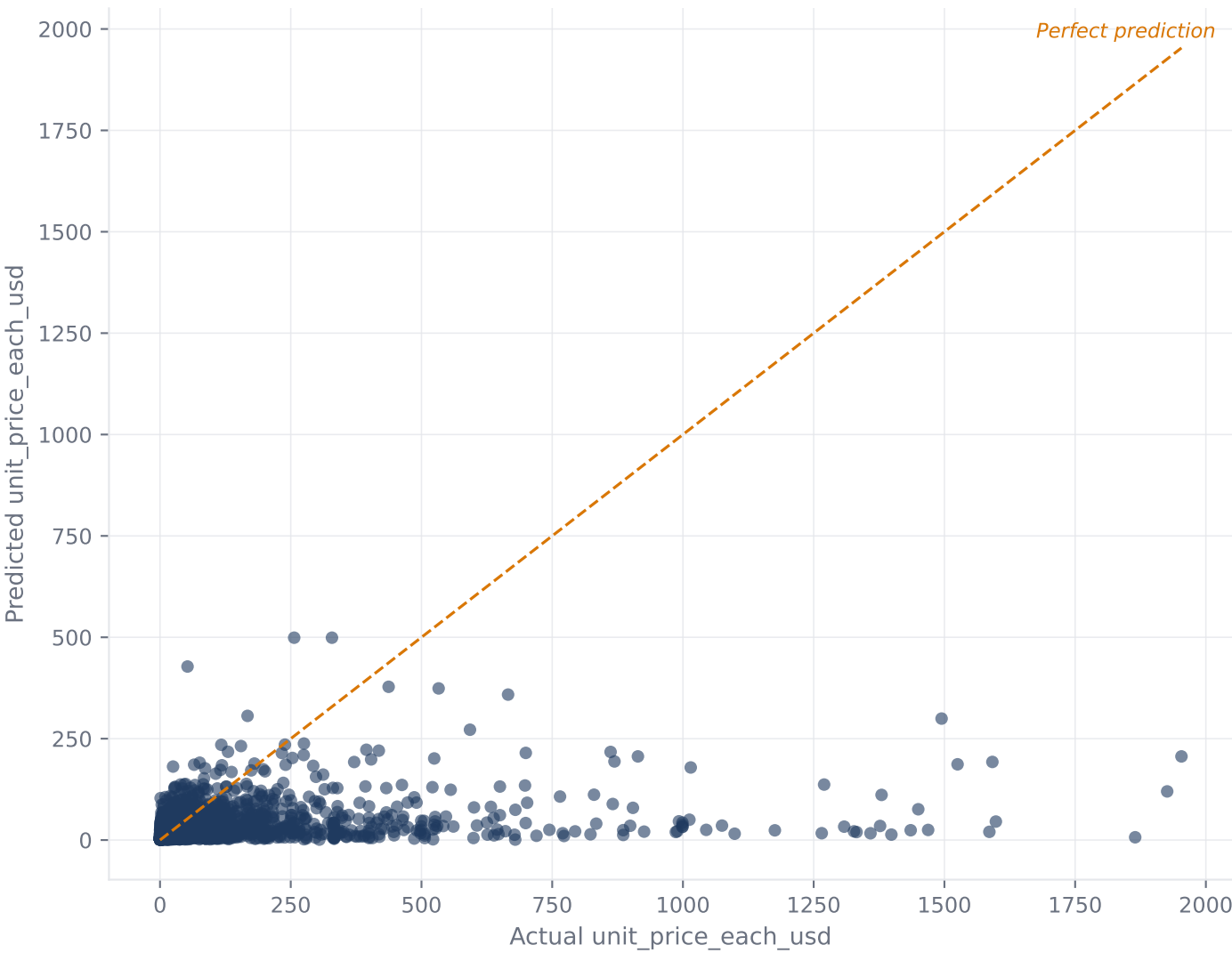
PROVENANCE

Model	xgboost
Target	unit_price_each_usd
Training rows	15,197
Holdout rows	3,800
Report date	2026-06-08 11:04:11

PERFORMANCE

Median % error	79.8%
Mean % error (MAPE)	258.1%
P90 % error	487.0%
MAE	53.32
RMSE	159.21
R ²	0.043

Predicted vs actual · holdout n = 3800



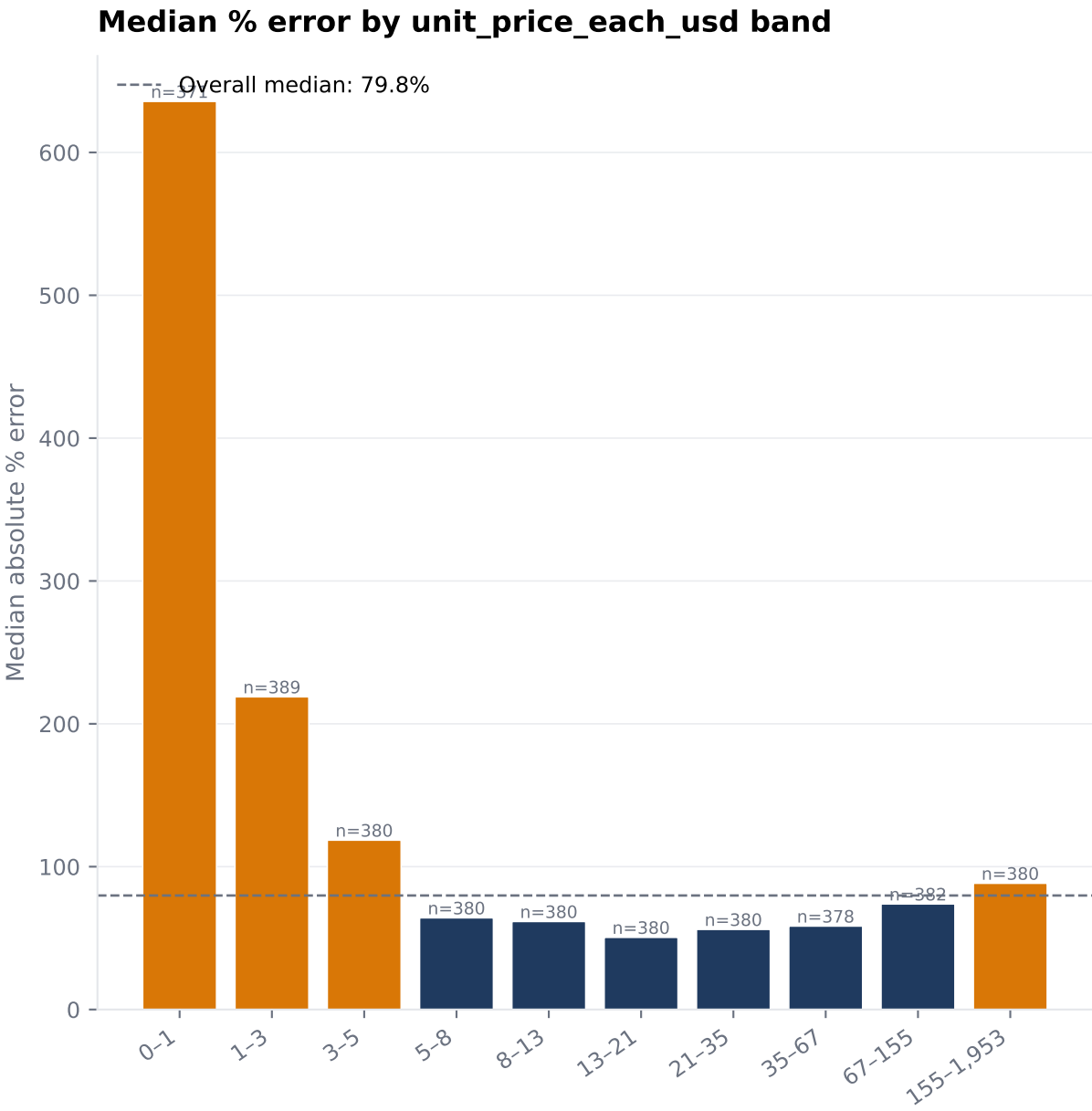
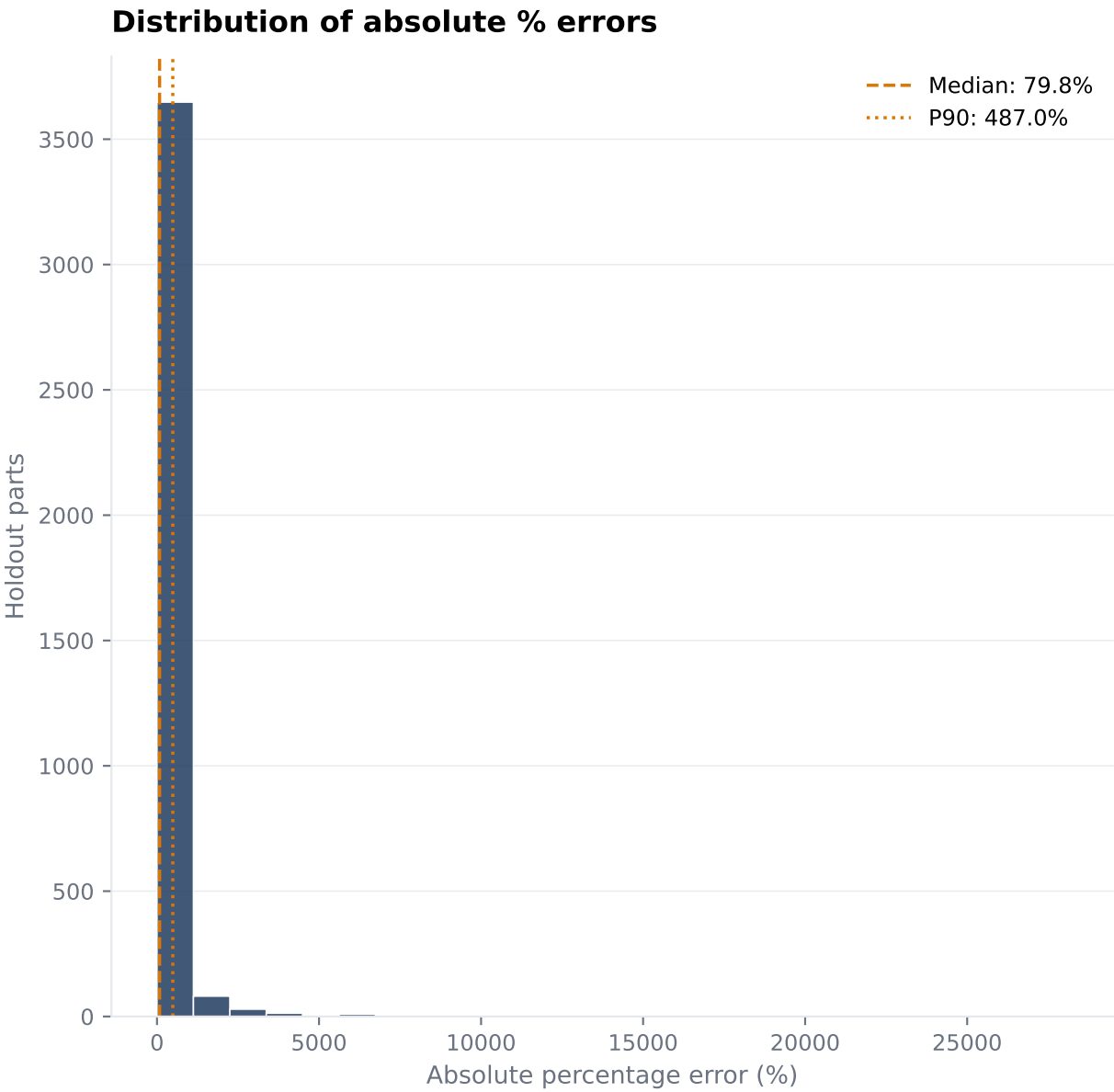
Definitions

Median = typical prediction error MAPE = average prediction error P90 = 9-in-10 worst-case error Holdout = parts the model never trained on

Error Distribution & Calibration

Where the model lands and which bands of the target it handles best

Accuracy is strongest in the 13-21 band (50.5% median error) and weakest in 0-1 (635.7%).



Feature Importance

Top 10 attributes the model relies on when estimating unit_price_each_usd

The top 3 features (thread_diameter_in, thread_series, material) explain 48% of the model's decisions.

